From Charminar to Cyberabad:

State Policy and Software Industry Development

in Andhra Pradesh

Abstract

In the 1990s, Andhra Pradesh was marked by agrarian distress, poverty, and, by the late 1990s, a crisis of farmers' suicides. Over the following decade, Hyderabad in erstwhile unified Andhra Pradesh emerged as one of India's leading software centres. This article examines how public policy contributed to that transformation. Drawing on forty-five semi-structured interviews with policymakers, IT workers, entrepreneurs, and industry intermediaries, alongside analysis of state ICT policy documents and secondary economic data, it traces the political economy of software industry development between 1999 and 2014. The article argues that the expansion of the IT sector cannot be explained by market forces alone. Rather, it was produced through the interaction of regional state policy, agrarian capital, private educational investment, and global outsourcing demand. Two linked policy mechanisms were central: the creation of technology-oriented urban infrastructure, including software technology parks and special economic zones; and the expansion of technical education that supplied labour to the industry. By examining software sector development at the subnational level, the article shows how a regional state can assemble labour, capital, and infrastructure in support of knowledge-based growth, while also reproducing spatial and social inequalities.

Keywords: Information Technology (IT), State Policy, Technical Education, Labour Formation, Andhra Pradesh

Introduction

The rise of the information technology (IT)[1] industry has been central to the making of erstwhile unified Andhra Pradesh as a model of post-liberalisation development. From the late 1990s onward, Hyderabad was promoted as a leading information technology destination, and the state government presented information technology as both an engine of growth and a tool of administrative modernisation (Mohanty et al., 2003). At the national level, the spread of computing and telecommunications had already come to be linked to a wider language of modernisation, efficiency, and self-reliance, with policymakers including Rajiv Gandhi arguing that information technology could contribute to social and economic development (Dedrick and Kraemer, 1993). In Andhra Pradesh, however, the rise of the IT sector was not simply a reflection of national trends. It was shaped by a regional political project that organised infrastructure, labour, and capital in support of IT sector growth.

This project took shape in a state marked by agrarian distress, uneven development, and pressure for economic restructuring. In the early 1990s, poverty remained high across both rural and urban Andhra Pradesh (Planning Commission, Government of India, 1997). By the 1990s, several regions of the state were experiencing deep agrarian stress, and farmers' suicides had become an increasingly visible political issue in districts such as Guntur, Prakasam, Srikakulam, and Rangareddy (Kumar, 2006; Vaditya, 2017). Balagopal (2004) describes this as a period of "poor growth of employment opportunities, and poor capital formation in agriculture". Following the 1991 liberalisation, Andhra Pradesh became the first state to obtain an independent World Bank loan under the AP Economic Restructuring Programme (Reddy and Prasad, 2014), a signal of both fiscal stress and external pressure for structural reform.

Yet within little more than a decade, Hyderabad in unified Andhra Pradesh had emerged as one of India's leading software hubs. This transformation was not automatic. Other southern states shared comparable endowments of technical labour in this period, yet none assembled the same configuration of infrastructure incentives and agrarian capital redirection that

[1]Information Technology (IT), Information and Communication Technology (ICT), and the software industry are used interchangeably in this article.

this article traces in AP (Kumar, 2009). A software industry required urban infrastructure, political brokerage, investor confidence, and a labour force with technical credentials. These conditions did not simply exist in place; they had to be assembled. This article therefore asks: how did state policy in erstwhile unified Andhra Pradesh shape the institutional conditions for the emergence and expansion of the software industry between 1999 and 2014?

Understanding this transformation matters for broader debates on development and structural change. The rapid expansion of India's software industry is often attributed to global outsourcing trends, the availability of inexpensive skilled labour, and the spread of information technologies. Yet such explanations risk obscuring the role of regional governments in creating the institutional conditions that allow new industries to emerge. In the case of Andhra Pradesh, the growth of the IT sector occurred alongside a broader shift toward a services-led development model centred on Hyderabad. Examining how the state facilitated this transition sheds light on the ways in which public policy can mobilise infrastructure, labour, and private capital to support the development of knowledge-based industries in emerging economies.

This article argues that the emergence of the software industry in Andhra Pradesh was actively shaped by public policy through two linked mechanisms. First, the state promoted the creation of technology-oriented urban infrastructure, including software technology parks and special economic zones, which provided firms with physical infrastructure and policy incentives. Second, it enabled the expansion of technical education, facilitating private investment in engineering colleges and training institutes that supplied labour to the industry. Together, these processes created the institutional conditions necessary for the rapid expansion of the IT sector.

The analysis focuses on the period from 1999 to 2014, spanning the introduction of key IT policies to the bifurcation of the unified Andhra Pradesh. The paper adopts a mixed-methods approach, combining analysis of government policy documents with forty-five semi-structured interviews conducted with policymakers, IT workers, entrepreneurs, and intermediaries, alongside secondary data and existing literature.

While there is a substantial literature on India's IT sector, fewer studies have examined how regional state policy organised infrastructure, labour formation, and private capital in support of software industry growth. By examining Andhra Pradesh, this article shows how a regional state can create and support knowledge-sector growth while also reproducing uneven development. In doing so, the paper contributes to debates on state-led restructuring, service-led growth, and the social foundations of post-liberalisation development.

Background: State Politics and the Genesis of the IT Industry

The Telugu Desam Party (TDP) government under Nara Chandrababu Naidu, which came to power in 1995, pursued a strategy of structural transformation oriented towards foreign investment, urban infrastructure, and professional middle-class growth — what Yeoh et al. (2004) describe as Singapore-inspired city-centred capitalism. A model that concentrated investment in urban enclaves, prioritises export-oriented services, and courts foreign capital through designated technology zones. As Damodaran (2018) documents, this produced rapid capital accumulation in Hyderabad and Visakhapatnam across sectors including information technology, pharmaceuticals, private education, healthcare, and real estate.

This strategy unfolded within a longer structural transformation of the state economy. Vakulabharanam and Motiram (2014) identify a transition beginning in the 1980s, during which services overtook agriculture as the dominant contributor to state gross domestic product (SGDP). Figure 1 plots the sectoral shares of SGDP and shows this longer movement towards a services-led economy.

After displacing N.T. Rama Rao within the TDP in 1995, Chandrababu Naidu served as chief minister from 1995 to 2004. His government placed far greater emphasis on urban growth, infrastructure, and investment promotion than on agrarian restructuring. Prepared with McKinsey & Company's assistance, *Vision 2020* was central to this reorientation. The document was less a technical plan than a political statement of intent: it signalled that the state's development ambitions were henceforth tied to technology, urban growth, and global

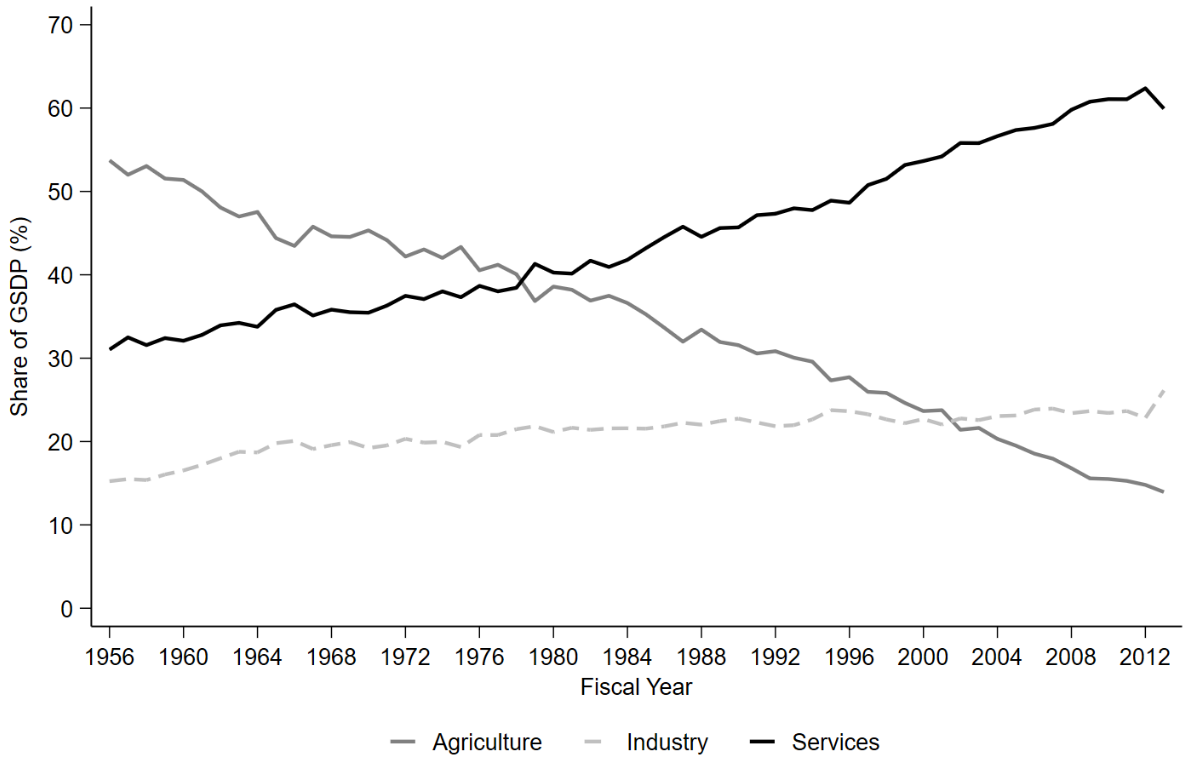

Figure 1: Sectoral Shares of GSDP, 1956–2014.

Source: Directorate of Economics and Statistics, Government of Andhra Pradesh.

integration rather than agrarian restructuring.

> 'Andhra Pradesh will move into the new millennium inspired by this vision and fortified by the knowledge that, implemented successfully, it will bring happiness, prosperity and fulfillment to its people' (Government of Andhra Pradesh, 1999, p. 338).

Bandyopadhyay (2001, p. 903) argues that in a setting marked by landlessness, exploitative tenancy, and entrenched rural poverty, large investments could produce weak social returns if the agrarian question remained unresolved. This criticism proved politically consequential: the TDP's development strategy generated visibility and investment in Hyderabad, but deepened the perception that the rural sector had been neglected amid worsening agrarian distress (Reddy and Prasad, 2014; Vaditya, 2017). The TDP lost power in 2004, with rural distress and perceptions of neglect forming an important part of the political context (Vakulabharanam and Motiram, 2014).

The Congress government under Y.S. Rajasekhara Reddy that followed adopted a different political style. It expanded welfare schemes such as Aarogyasri and educational subsidies while continuing to support private capital and urban growth (Reddy, 2002; Suri, 2004). The two governments differed in emphasis but not in the broader direction of accumulation: both relied on active state intervention to support private investment, though through different instruments — TDP through infrastructure and foreign investment promotion, Congress through the expansion of private technical education.

The cumulative effect of both policy regimes was to position Hyderabad as one of India's major centres for IT and IT-enabled services (Upadhya, 2011). Figure 2 shows the administrative map of Andhra Pradesh prior to bifurcation. This territorial frame is significant: the transformation examined here was organised at the scale of the unified state, but its gains were concentrated overwhelmingly in and around Hyderabad.

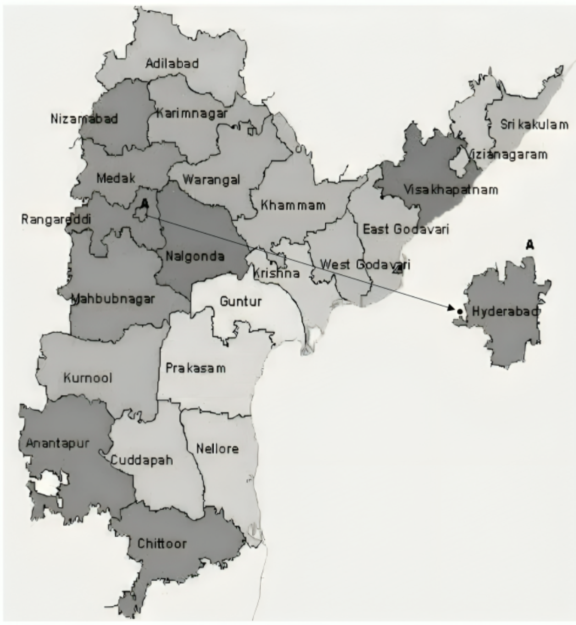

Figure 2: Administrative Division Map of Andhra Pradesh (Census 2011).

Hitec City[2] became the most visible spatial expression of this project. By the late 1990s, Hyderabad offered many of the conditions sought by software firms: improving telecommunications, dependable power in key enclaves, growing urban infrastructure, and an expanding pool of technically trained labour (Popham, 1999). The city was promoted as an investment

[2]HITEC stands for Hyderabad Information Technology and Engineering Consultancy City.

destination not only because of these material advantages, but because the state actively assembled them through land allocation, infrastructure provision, and political brokerage. Chandrababu Naidu's personal role in attracting firms, including through direct engagement with global technology executives, was widely noted at the time (Biswas, 2004). Over time, Hyderabad's public image shifted from Charminar to Cyberabad — from a city associated with its historical urban core to one increasingly defined by its technology economy (Ramachandraiah et al., 2008).

The scale and pace of growth were substantial. Software and software services exports increased from US$7.1 million in 1994 to US$2,828.3 million in 2005 (Kumar, 2009, p. 107). Figure 3 plots the expansion of Software Technology Park (STP) units alongside IT exports. The sharp increase in STP units from the late 1990s coincided with rapid export growth. By the time of bifurcation in 2014, the sector had grown to approximately 1,300 IT units employing 3.7 lakh professionals, with an annual export turnover of Rs 68,258 crores (Ramachandraiah, 2016). These aggregate figures, however, capture only the scale of expansion, not its distribution. The gains were highly uneven in both geography and social reach, an unevenness that was not incidental to the model but structural to how it was designed.

Methodology

The article adopts a mixed-methods design combining analysis of government policy documents, semi-structured interviews, and secondary economic data. This approach is used to examine both the institutional mechanisms through which the software industry was promoted and the ways in which these mechanisms were experienced and mediated by different actors. Rather than treating policy as a set of formal instruments alone, the analysis links policy design to processes of labour formation, capital mobilisation, and spatial concentration.

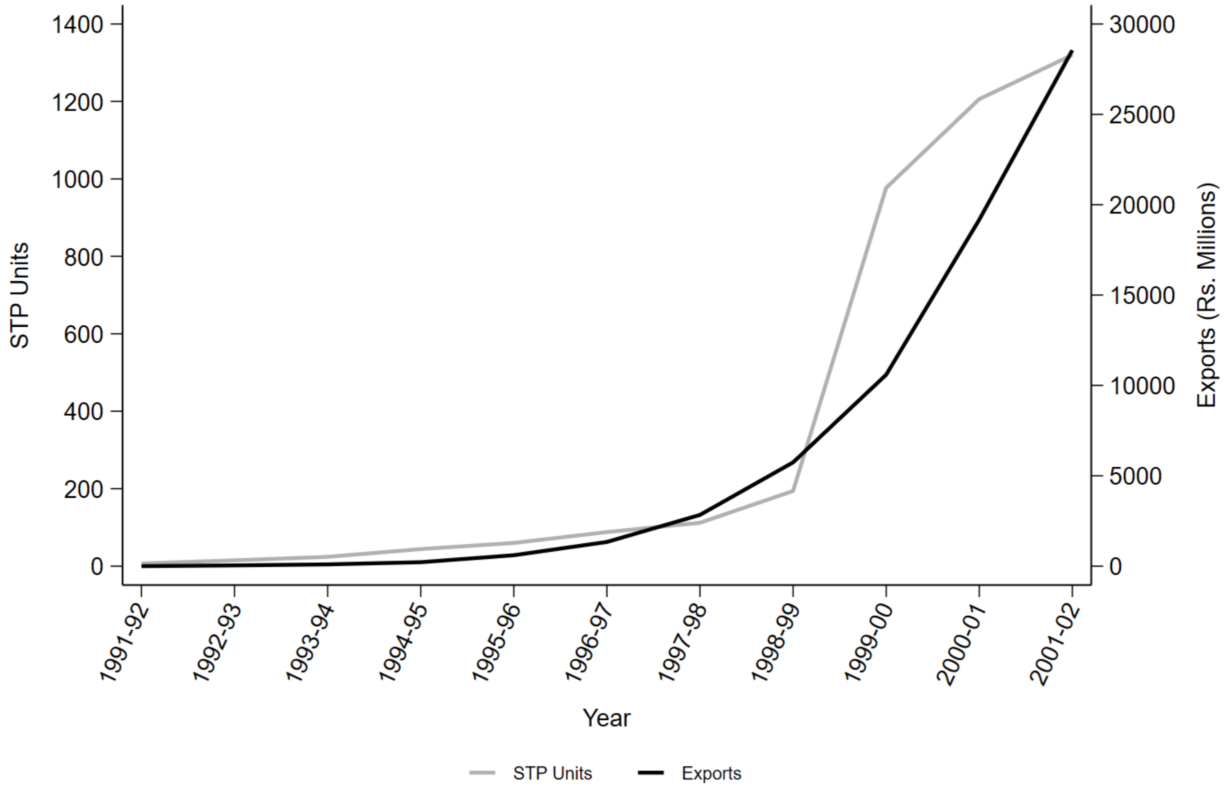

Figure 3: Expansion of Software Technology Parks and IT Exports in AP, 1991–2002

Source: Software Technology Parks of India (Hyderabad); www.ap-it.com; Ramachandraiah (2003).

Fieldwork

Fieldwork was conducted in Hyderabad in 2018, primarily at Cyber Towers and Nanakramguda in the Financial District, the core sites of the city's IT economy and the locations where the spatial concentration of the industry is most visible. Forty-five semi-structured interviews were carried out with a cross-section of industry stakeholders. Respondents comprised engineers, human resource personnel, technicians, and managers (33); industry intermediaries[3] (5); government officials (2); and AP/Telangana-native IT entrepreneurs (5). Informed consent was obtained from all respondents and names have been anonymised throughout.

Purposive sampling was used to capture variation across company types, including software design firms, IT-enabled services (ITES)[4], and customer care operations.[5] Within the sample

[3]Industry intermediaries facilitate interactions between firms, clients, and regulatory processes, including licensing, compliance, and service coordination. Several respondents used the term 'middlemen' to describe this role.

[4]ITES firms are primarily engaged in coding, testing, and related technical support functions.

[5]These firms may be service-based or product-based.

of engineers and managers, respondents included 6 managers and executives, 14 consultants and team leads, 5 human resource personnel, and 8 coders and testers. The five AP/Telangana-native entrepreneurs were all male, held postgraduate qualifications, and had established their firms using capital accumulated either from agriculture or from prior employment in IT firms abroad — a pattern that reflects the broader agrarian and diasporic origins of capital discussed in later sections.

The interviews are used to trace how policy was interpreted, implemented, and experienced across different positions within the industry. They are not intended to be representative of the sector as a whole. Open-ended questions allowed respondents to speak at length about working conditions, family background, gender and caste in the workplace, modes of payment, and perceptions of industry change over time.[6] Interviews were conducted face-to-face, by telephone, and by email. While the sample size does not support statistical generalisation, it provides qualitative insight into how labour and capital mobilisation operated in practice and how different actors understood the role of the state in shaping the industry.

Interview transcripts and notes were coded thematically, with attention to recurring patterns in respondents' perceptions of state policy, labour formation, and industry structure. Policy documents were analysed to identify the instruments and stated objectives of successive state interventions, and triangulated against interview evidence and secondary economic data.

Policy Analysis

The paper analyses ICT and related policy documents issued by the Government of Andhra Pradesh (GoAP) between 1999 and 2014, covering the period from the first state IT policy to bifurcation.

Table 1 lists the primary and secondary policies examined. Central government policies are included where they shaped the state-level policy environment, though the aim is not to isolate the effect of any single policy. Instead, the analysis treats policy as a cumulative process, examining how successive interventions in infrastructure, education, governance, and

[6] Questionnaires are available upon request.

Table 1: Policies with a focus on the IT industry in Andhra Pradesh

Policies with primary focus on the IT industry in Andhra Pradesh	Policies that indirectly helped the IT industry in Andhra Pradesh
Vision (2020)	EXIM policies (1985, 1992-97, 2002-07)
ICT press release (2002)	Industrial policy (1991)
IT-ITes promotions (2002-2005)	Labour Reforms policy
ICT policy (2005-10)	SEZ policy (2005)
Notification of ITIR (May 29th 2008)	Science, Technology and Innovation Policy (2013)
ICT policy press release (2010-15)	Foreign Trade Policies (2001-14, 2015-20)
ICT policy (2014)	Technical Education policies (1968, 1986, 1992, 2015)

Source: Prepared by the author on the basis of information obtained from MOSPI, RBI, GoAP, NASSCOM, and www.ap-it.com.

investment promotion combined to shape the conditions for software industry growth. This approach allows the paper to identify the institutional logic linking policy measures to labour supply, capital flows, and the spatial concentration of the sector.

State Policy, Labour Formation, and the Political Economy of IT Growth

In many late-developing economies, structural transformation is actively mediated by the state rather than left to market coordination alone (Amsden, 2007). Andhra Pradesh followed this pattern, but with a distinctive trajectory: instead of moving from agriculture to manufacturing and then to services, the state attempted to channel labour and capital more directly from agriculture into services — what Miller (2001) describes as leapfrogging into a knowledge economy. This trajectory, however, was neither frictionless nor socially neutral. It depended on a specific regional configuration of agrarian surplus, caste power, urban land development, and technical education.

The shift was made possible in part by an earlier agrarian transformation. The green revolution raised agricultural incomes in parts of the state and produced a surplus-holding rural class with the capacity to invest outside agriculture (Upadhya, 2016). As Damodaran (2018) shows, rural capitalists from coastal Andhra redirected this surplus into urban sectors such as private education, healthcare, and real estate. The dominant castes that controlled agricultural land and capital — Kammas, Reddys, Kapus, Rajus, and Velamas — were also prominent among the owners of private technical education institutions, hospitals, and, in some cases, IT firms. Two early examples illustrate the broader pattern: Sri Chaitanya Techno Schools, established in Hyderabad in 1986 by B.S. Rao from Tadigadapa near Vijayawada, and Satyam Computers, founded in 1987 by Byrraju Ramalinga Raju from Bhimavaram. Public policy did not generate this capital ex nihilo. Rather, it made the redirection of agrarian surplus into urban knowledge-sector institutions more viable and more profitable.

The preconditions for an IT industry in Hyderabad were therefore already partially in place before the major policy push. As a Professor from the Centre for Economic and Social Studies explained during fieldwork:

> ‘The software industry started in Hyderabad in early 1990s along with the other major cities (except Bengaluru). Software Technology Parks of India (STPI), Hyderabad was set up in 1992 . . . Hyderabad was already a 4 million-plus city by then. It had a number of educational institutions including a few universities, and a number of public sector undertakings. The availability of required manpower was not a problem. It just needed a push . . . Any push was a welcome move in those days. To begin with, probably, the push was for infrastructure, a place with plug-and-operate facilities from where new companies can make a beginning. Later on it was a policy push with providing an enabling environment and incentives.’
> — *Professor, Centre for Economic and Social Studies (CESS), Hyderabad*

What followed was precisely such a push. State policy worked through two closely related mechanisms: first, the construction of urban infrastructure and incentive regimes that lowered the cost and risk of investment; second, the expansion of technical education and training that generated a large labour pool for the industry. The sections below examine each mechanism in turn.

Infrastructure and Investment Incentives

In 1999, with consultancy support from McKinsey & Company, the TDP government formulated *Vision 2020*, which provided the strategic framework for subsequent IT interventions. The policy sought to attract major global firms to Andhra Pradesh, develop international alliances, create venture capital mechanisms, and build Hitec City in Hyderabad as a globally competitive technology enclave (Government of Andhra Pradesh, 1999). Drawing on examples such as Research Triangle Park in North Carolina and Hsinchu Science Park in Taiwan, the policy presented 'knowledge corridors' and 'knowledge workers' as the basis of a new regional growth model.[7]

The principal secretary of the IT Department in the combined state of AP described the reasoning behind this strategy:

> 'The initiation of the promotion of IT as an industry with a great potential for AP happened during 1996–99, with the vision of the then Chief Minister of AP. The hypothesis was that AP and Hyderabad had significant advantages in terms of infrastructure, human resources and pro-active government, which could be leveraged effectively to make rapid strides for the growth of IT Industry.' — *Principal Secretary of IT Department, combined AP, 1999–2004 (Field Notes)*

The ICT Policy 2002–05 translated this strategy into concrete incentives: exemptions from statutory power cuts, subsidies for pollution-control investments, reimbursement of registration fees and stamp duty, and relaxation of zoning regulations for IT firms (Government of Andhra Pradesh, 2002). IT and ITES Special Economic Zones (SEZs) were promoted in Vijayawada, Tirupati, Kakinada, and Warangal, while Software Technology Parks offered export-oriented firms ready-to-use infrastructure and fiscal advantages (Government of Andhra Pradesh, 2005a,b). Together, these measures reduced entry barriers and concentrated investment in designated technology spaces. As Figure 3 suggests, the rapid rise in STP units from the late 1990s coincided with accelerating export growth.

[7] Vision 2020 defines a knowledge worker as 'an individual with specialised knowledge and skills in fields like software, medicine, accounting or engineering. The more an economy can utilise and attract knowledge workers, the faster is its development' (Government of Andhra Pradesh, 1999, p. 286).

The 2002–05 policy recorded the scale of the expansion: software exports rose from Rs 574 crores in 1998–99 to Rs 2,855 crores in 2001–02, while employment increased from 12,000 to 64,000 (Government of Andhra Pradesh, 2002). These figures should not be read as the product of state policy alone. National liberalisation, telecom reforms, and expanding global outsourcing demand were also crucial. But state policy was a decisive regional mediator: it translated these wider shifts into a territorially concentrated growth process centred on Hyderabad.

A parallel dimension of this infrastructure push was e-governance. The Chief Minister's executive information system, operational from 1998, provided decision-makers with real-time data on infrastructure and public services. The eSeva programme, introduced in 1999, digitised bill payments, tax applications, and licences (Government of Andhra Pradesh, 2002, 2005a). These initiatives did more than modernise administration. They also signalled state capacity to investors and created internal demand for IT-enabled systems within government itself. Table 2 summarises the policy sequence and its principal instruments.

Table 2: Summary of Policies

Time Period	Policies	Summary
1999	Vision, 2020	Formation of software technology parks, creation of SEZs and cyber towers
2002-2005	IT-ITes promotions, 2002-2005	Setting up software parks, digitising administrative data
2005-2010	ICT policy, 2005- 10	Promoting educational institutions - engineering colleges and coaching centres.
2010-2015	ICT policy press release, 2010-15	Promotion of entrepreneurship in the state. Focus on providing more jobs. Push for startups and planning to increase special economic zones (SEZs).
2014	ICT policy, 2014	Providing more IT employment and subsidies for women and SC/ST entrepreneurs preferably in rural areas.

Yet this developmental strategy had clear political limits. *Vision 2020* was criticised for opaque assumptions and technocratic optimism (RV, 1999). As Suri (2004) notes, voters in the 2004 National Election Study judged Chandrababu Naidu poorly on irrigation and employment. In a state marked by agrarian mobilisation, Naxalite insurgency, and farmers' suicides (Kumar, 2006; Ravikanti, 1995; Simdhari, 1997), an urban-centred IT strategy could not by itself secure broad political legitimacy. The software boom was therefore not simply a story of successful growth policy. It was also part of a wider reordering of public priorities that generated exclusion as well as accumulation.

Technical Education and Labour Supply

The second major mechanism was the expansion of technical education. Across successive ICT policies, engineering colleges, computer training institutes, and coaching centres were treated as the principal means of generating an employable labour force for the software industry. As a professor from IIIT Hyderabad noted during fieldwork, Andhra Pradesh was widely seen as a state with a large pool of technically trainable labour.

The government supported this expansion mainly by facilitating private investment rather than by building a public system at scale. Engineering courses and syllabi were revised to align with industry expectations; IT training institutes were encouraged; online certification was promoted; and internships were made mandatory (Government of Andhra Pradesh, 2002, 2014). Elite institutions such as the Indian School of Business (ISB) and IIIT Hyderabad represented the upper end of this system, while hundreds of private engineering colleges and computer coaching centres supplied volume. In practice, this was a labour-production strategy: the aim was not only to educate, but to generate a large, flexible, and relatively low-cost workforce.

The most visible result was the rapid increase in engineering and technology colleges. Figure 4 shows that by 2007–08 Andhra Pradesh had 319 AICTE-approved engineering institutions — more than Karnataka and far more than Kerala. This concentration reflected not only rising demand, but also a policy environment unusually favourable to private educational capital.

This expansion was financed largely by private capital, but it rested on the same social base that underpinned the wider urban transformation. As Xiang (2007, p. 31) notes, surplus capital and labour from the agricultural sector were central to the rise of these higher education institutions. Computer coaching centres — what Xiang terms 'body shops' — also widened the labour pipeline by training students outside formal engineering streams, including graduates from arts and commerce backgrounds. Technical education thus became both an industry input and a social aspiration. In Xiang's formulation, Telugu families came to 'produce IT people as a family business', with implications that extended into household strategies, migration, and marriage markets (Xiang, 2007, p. 37).

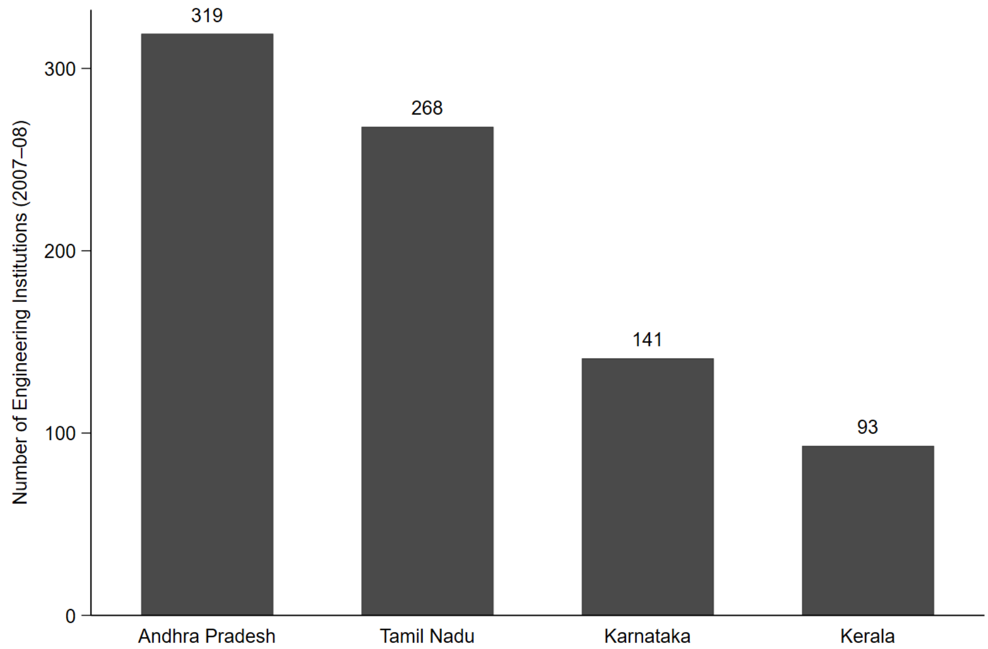

Figure 4: Engineering Institutions in Major South Indian States, 2007–08. Source: AICTE Handbook for Approval Process (2008).

The principal secretary of the IT Department described the state's view of the process in the following terms:

> 'Since high-quality high-volume human resources were key to success, adequate efforts were made to increase the number of institutions of technical training, to promote quality of education and to establish Knowledge Centres in various engineering colleges. The emergence of IIIT Hyderabad as a premier IT institution is an example of such efforts.' — *Principal Secretary of IT Department, combined AP, 1999–2004 (Field Notes)*

The expansion of technical education also produced a wider cultural shift. Upadhya (2014) documents how families across Andhra Pradesh oriented schooling and social aspiration towards becoming a 'techie', often from an early age through competitive examination preparation. As one informant, Gopal, put it during fieldwork: 'Anyone in the street who puts his shirt inside his trousers is an IT professional.' The remark is humorous, but it points to how fully IT employment had entered popular aspiration and social identity.

Access to this aspiration, however, was highly unequal. Gundemeda (2014) shows that caste, class, parental education, income, and social networks shaped who entered IT education and who could convert credentials into employment. In Seemandhra and Rayalaseema, IT aspirants came predominantly from upper castes; in Telangana, they were more likely to come from backward communities. These differences mattered even more when labour markets tightened. Students with stronger social networks retained confidence in their prospects, while others faced a credential-heavy but effectively closed labour market (Gundemeda, 2014, p. 325). Jeffrey et al. (2007) describe a related outcome: graduates who were over-credentialised for non-IT work but under-prepared for the sector for which they had trained. A professor from IIIT Hyderabad offered a pointed critique of this dynamic:

> 'Coaching culture is the single biggest problem. Almost every 11th and 12th student in the country is preparing for entrance exams. They are just filling MCQs. There is no knowledge, no inspiration, no motivation, no stimulation . . . They are throwing away two years of development for a whole generation. Every one need not prepare for IIT. One-third of engineering colleges in the state are good — they are enough.' — *Professor, IIIT Hyderabad (Field Notes)*

The Structure of IT Work

The labour force generated through this system was large and technically credentialled, but much of it was channelled into lower-value back-end work. The export orientation of the industry shaped not only the volume of employment but also the organisation of work itself. A software engineer from Rajahmundry working at Infosys in Hitec City was, in practice, often working for a foreign client such as Goldman Sachs, whose requirements shaped team structures, deadlines, and project tasks. This client-driven organisation of production had consequences for mobility and power within firms. As one informant, Suchitra, observed, women were included in project teams to satisfy client expectations of diversity, yet were often excluded from leadership roles because they were not assumed to be continuously available for client-driven schedules.

A vice president for technology management at a major conglomerate described the structural consequence of this growth pattern as follows:

> '95 per cent of IT is concentrated in Hyderabad. There is absolutely nothing IT in Andhra Pradesh … There are two kinds of jobs in IT: the developer community and support services. Support services have seen phenomenal growth. Most of those jobs are low paid. The developer community or any knowledge base in IT has not grown substantially … Only one in ten jobs contribute to the IT base. The base is not growing at the right pace.' — *Vice President, Technology Management (Field Notes)*

This observation points to a basic limit of the model. State policy succeeded in generating a large workforce for coding, testing, customer care, and ITES work, but it did not produce equivalent growth in higher-value product development or knowledge-intensive design. Kumar (2009) links this outcome to the industry's export orientation from the outset, which weakened linkages with local institutions and concentrated benefits in larger firms. As Teja, an industry intermediary, noted: 'The government was promoting only the bigger companies by providing them labour and infrastructure. There were so many small companies that came up but remained small.'

The result was an industry that was impressive in aggregate yet structurally constrained: geographically concentrated in Hyderabad, dependent on foreign clients, stratified by caste and gender, and oriented more towards volume than technological depth.

Discussion and Conclusion

This article set out to explain how public policy in erstwhile unified Andhra Pradesh promoted the growth of the software industry between 1999 and 2014. The evidence from policy documents, fieldwork, and secondary data points to two linked mechanisms. First, the state created technology-oriented urban infrastructure — software technology parks, special economic zones, and the incentive environment of Hitec City — that lowered the cost and risk of locating in Hyderabad for foreign IT firms. Second, it enabled the rapid expansion of private technical education, producing a large pool of engineering graduates and computer-trained workers that the industry could absorb. Neither mechanism alone was sufficient; together, they created the institutional conditions for a software boom that market forces alone would not have produced.

This finding contributes to broader debates on late industrialisation and state-led development. The AP case supports (Amsden, 2007) argument that in late-industrialising economies, the state must actively construct the conditions for new industries rather than waiting for them to emerge spontaneously. It also illustrates (Miller, 2001) leapfrogging argument in practice: AP moved directly from an agrarian economy into services, bypassing manufacturing, through deliberate policy choices rather than gradual structural change. What the case adds to these frameworks is the specific role of agrarian capital — surplus held by dominant-caste rural elites redirected into urban educational infrastructure — as the financial substrate that made the labour supply side of this model possible. The state did not fund this capital formation directly; it created the policy environment in which redirecting agrarian surplus into private engineering colleges became rational and profitable.

The model had significant structural limits, however, and these deserve as much attention as its achievements. Growth was geographically concentrated in Hyderabad to a degree that excluded the rest of the state. The uneven spatial distribution of IT investment was not an oversight but a structural feature of how the model was designed: infrastructure and incentives were concentrated in one city, and the labour that flowed towards it came disproportionately from across AP rather than from Hyderabad itself. This concentration was a contributing factor in the political pressures that led to bifurcation in 2014, when Telangana's claim that the benefits of IT-led growth had bypassed the region gained sufficient political force to result in the division of the state (Vakulabharanam and Motiram, 2014).

The industry that emerged was also structurally oriented towards back-end operations rather than knowledge-intensive product and service design. As one informant noted, working at Infosys meant working for Goldman Sachs — client demands, not firm strategy, determined what work was done and how teams were organised. Jagath, an IT engineer, described his experience: 'I have been working in the industry for three months and all I did was cut, copy and paste the same code for different projects, nine to five, every working day.' This export dependence weakened local linkages, concentrated benefits in larger firms, and left the industry vulnerable to global demand fluctuations (Kumar, 2009). The overproduction of engineers compounded this: the expansion of private colleges generated more graduates than the industry could absorb at the skill level it required, producing a credentialised but underemployed labour surplus (Gundemeda, 2014; Jeffrey et al., 2007).

Access to the industry's benefits was further structured by caste and gender. IT employment was not equally available to all who sought it. Caste, class, parental education, and social networks determined who accessed quality technical education and who converted credentials into employment (Gundemeda, 2014). Women entered the workforce in significant numbers but were systematically excluded from positions of power. The aspiration that made the 'techie' identity so powerful was real; the distribution of that aspiration's rewards was not.

After bifurcation, the successor state of Andhra Pradesh continued to pursue IT-led development through Vision 2029, which proposed IT hubs in Visakhapatnam, Vijayawada, Kakinada, and Anantapur, and a cyber towers equivalent in Vizag. The post-bifurcation policy also introduced new provisions — exemptions from labour laws permitting three-shift operations, minimum employment requirements per acre of land allocated, and targeted subsidies for SC, ST, and women entrepreneurs. Whether these adjustments address the structural limits of the earlier model, or reproduce its spatial and social concentrations in a new geography, remains to be seen.

The directions for future research follow from this analysis. First, the post-bifurcation divergence between Telangana and Andhra Pradesh — where Hyderabad remained with Telangana — offers a natural experiment in what happens when an IT hub is separated from the agrarian hinterland that supplied its labour and capital. Second, the caste dimension of IT access and mobility, which this paper raises but cannot fully resolve, warrants dedicated research, particularly given the differential patterns across Seemandhra, Rayalaseema, and Telangana.

The broader lesson of the AP case is not that IT-led development is replicable wherever a government is sufficiently proactive. It is that the conditions that made it possible in AP — a specific configuration of agrarian capital, state capacity, aspirational labour, and global outsourcing demand — were historically particular. Understanding how they were assembled, and who was included and excluded in the process, is what this article has attempted to do.

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